AI / ML / Python / DSA Trainer Evaluation - Answer Bank

Python

Q1. Difference between list and tuple?

Answer: Lists are mutable and use square brackets []. Tuples are immutable and use parentheses ().

Q2. Difference between == and is?

Answer: == compares values, while 'is' compares memory locations (object identity).

Q3. What are generators?

Answer: Generators produce values lazily using yield, saving memory.

Q4. Why did x change when y.append(4) was used?

Answer: Both x and y reference the same list object.

Q5. Why is default mutable argument dangerous?

Answer: The same list object is reused across function calls.

Q6. How would you optimize a slow Python script?

Answer: Use efficient algorithms, profiling, vectorization, caching, and optimized libraries.

DSA

Q1. Difference between array and linked list?

Answer: Arrays use contiguous memory and allow fast indexing. Linked lists use nodes with pointers and allow easier insertion/deletion.

Q2. What is recursion?

Answer: A function calling itself until a base condition is met.

Q3. Why does arr[3] give error?

Answer: Index out of range because valid indices are 0–2.

Q4. What is dynamic programming?

Answer: An optimization technique storing repeated subproblem results.

Q5. When should recursion be avoided?

Answer: For deep calls that may cause stack overflow or when iterative solutions are simpler.

EDA / Data Analysis

Q1. What is EDA?

Answer: Exploratory Data Analysis is the process of analyzing and visualizing datasets before modeling.

Q2. Difference between mean and median?

Answer: Mean is average value; median is middle value after sorting.

Q3. How do you detect outliers?

Answer: Using boxplots, IQR method, or z-score.

Q4. Why is df['salary'].mean incorrect?

Answer: mean is a method and should be called using mean().

Q5. Why is visualization important?

Answer: It helps identify patterns, trends, and anomalies quickly.

Machine Learning

Q1. Difference between supervised and unsupervised learning?

Answer: Supervised learning uses labeled data; unsupervised learning finds hidden patterns in unlabeled data.

Q2. What is overfitting?

Answer: Model performs well on training data but poorly on unseen data.

Q3. What is train-test split?

Answer: Dividing data into training and testing sets for evaluation.

Q4. Why is accuracy misleading sometimes?

Answer: In imbalanced datasets, high accuracy may still fail on minority classes.

Q5. How does Random Forest work?

Answer: It combines multiple decision trees and averages their predictions.

Deep Learning

Q1. What is a neural network?

Answer: A layered computational model inspired by the human brain.

Q2. What is backpropagation?

Answer: An algorithm that updates weights using gradients to reduce error.

Q3. Why is ReLU popular?

Answer: It is computationally efficient and reduces vanishing gradient problems.

Q4. Why do large models require GPUs?

Answer: GPUs perform parallel computations efficiently for matrix operations.

Q5. When should ML be preferred over DL?

Answer: When data is limited or interpretability is important.

Logical & Critical Thinking

Q1. How would you handle imbalanced datasets?

Answer: Use oversampling, undersampling, class weights, or suitable metrics like F1-score.

Q2. Can correlation imply causation?

Answer: No. Correlation only indicates relationship, not cause-effect.

Q3. Why should test data never be used during training?

Answer: It causes data leakage and unrealistic performance estimates.

Q4. Why is deep learning data-hungry?

Answer: Deep models contain many parameters and require large datasets to generalize well.

Trainer Evaluation Rubric

Skill	Marks
Concept Clarity	20
Coding Ability	20
Debugging	20
Problem Solving	20
Communication	10
Optimization Thinking	10